

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	3	9	17

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

consider $\hat{y}_i = \hat{\theta}_1 x_{i1} + \hat{\theta}_2 x_{i2}$

so the given problem can be made as

$$Y = X\theta$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \epsilon$$

for least squares estimate we minimize error

$$\rightarrow \min \sum_{i=1}^3 \epsilon_i^2 \rightarrow \sum_{i=1}^3 (Y_i - \theta_1 x_{i1} - \theta_2 x_{i2})^2$$

$$\min (Y - X\theta)^T (Y - X\theta)$$

so we differentiate w.r.t θ and finally get

$$\theta = (X^T X)^{-1} X^T Y$$

$$\text{so } X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} \rightarrow \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{aligned}
 (X^T X)^T X^T Y &= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
 &= \frac{1}{9} \begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
 &= \begin{bmatrix} \frac{y_1 + y_3}{3} \\ \frac{y_1 - y_2}{3} \end{bmatrix}
 \end{aligned}$$

∴ the least square estimates of θ_1 and θ_2 are given by $\rightarrow$

$$\hat{\theta}_1 = \frac{y_1 + y_3}{3}$$

$$\hat{\theta}_2 = \frac{y_1 - y_2}{3}$$

* proof for $\hat{\theta} = (X^T X)^T X^T Y$

so we minimize $(Y - X\hat{\theta})^T (Y - X\hat{\theta})$

$$\rightarrow (Y^T - \hat{\theta}^T X^T)(Y - X\hat{\theta}) = Y^T Y + \hat{\theta}^T X^T X \hat{\theta} - \hat{\theta}^T X^T Y - Y^T X \hat{\theta}$$

differentiate w.r.t θ and equating to 0
we get

$$(X^T X + X^T X) \hat{\theta} - 2X^T Y = 0$$

$$\rightarrow 2X^T X \hat{\theta} = 2X^T Y \rightarrow \underline{\underline{\hat{\theta} = (X^T X)^T X^T Y}}$$

05

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $X = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of X .

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

a) it can be seen that

$$\text{Var}(X_1) = \text{Var} U_1 = \frac{1}{12} \quad (U_1 \sim U(0,1) \text{ so } \text{Var} U_1 = \frac{(1-0)^2}{12})$$

$$\text{Var}(X_2) = \text{Var} U_2 = \frac{1}{12}$$

$$\text{since } \text{Var}(X_3) = \text{Var} U_1 + \text{Var} U_2 = \frac{1}{12} + \frac{1}{12} = \frac{1}{6} \quad (\text{because } U_1 \text{ and } U_2 \text{ are independent})$$

$$\text{Corr}(X_1, X_2) = \frac{\text{Cov}(U_1, U_2)}{\sqrt{\text{Var} U_1 \text{Var} U_2}} = 0$$

$$\begin{aligned} \text{Corr}(X_1, X_3) &= \frac{\text{Cov}(U_1, U_1 + U_2)}{\sqrt{\text{Var} X_1 \text{Var} X_3}} = \frac{\text{Var} U_1}{\sqrt{\text{Var} X_1 \text{Var} X_3}} \\ &= \frac{1/12}{\sqrt{1/12 \times 1/6}} = \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{Corr}(X_2, X_3) &= \frac{\text{Cov}(U_2, U_1 + U_2)}{\sqrt{\text{Var} X_2 \text{Var} X_3}} \\ &= \frac{\text{Var} U_2}{\sqrt{\text{Var} X_2 \text{Var} X_3}} = \frac{1/12}{\sqrt{1/12 \times 1/6}} = \frac{1}{\sqrt{2}} \end{aligned}$$

so correlation matrix of $X = (X_1, X_2, X_3)'$ is

$$\textcircled{03} \quad \rho = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0.7071 \\ 0 & 1 & 0.7071 \\ 0.7071 & 0.7071 & 1 \end{bmatrix}$$

b) ~~no correlation~~ ^{principal} component is given by $\rightarrow$

$\lambda_1 \rightarrow$ eigenvalue of $\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2}$
 which is $\Sigma_{11}^{-1/2} = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1/2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\Sigma_{22}^{-1} = [1]^{-1} = [1]$

$\Sigma_{21} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ $\Sigma_{12} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

~~$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$~~ it becomes $\Sigma_{12} \Sigma_{21} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
 $= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$

~~this matrix has the eigen values $\rightarrow 0$ and 1~~

~~so $\lambda_1 = 1$ so $\lambda^* = \sqrt{\lambda_1} = 1$
 with corresponding eigenvector $\rightarrow \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$~~

~~so Principal component $\rightarrow$~~

~~$U \rightarrow e_1' \Sigma_{11}^{-1/2} X^{(1)} = \frac{X_1}{\sqrt{2}} + \frac{X_2}{\sqrt{2}}$~~

~~$U \rightarrow e_1' \Sigma_{22}^{-1/2} X^{(2)} = X_3$~~

at back on page no. 9

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) i) in complete linkage, distance between 2 clusters is given as
 $d(UV, W) = \max(d(U, W), d(V, W))$

$$\begin{array}{c}
 \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & & & \\ \textcircled{1} & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \xrightarrow{\text{join 1, 2 into a cluster}} \begin{array}{c} (12) \\ 3 \\ 4 \end{array} \begin{bmatrix} (12) & 3 & 4 \\ 0 & & \\ 11 & 0 & \\ 5 & \textcircled{3.5} & 0 \end{bmatrix}
 \end{array}$$

$$\begin{aligned}
 d((12), 3) &= \max(d(1, 3), d(2, 3)) \\
 &= \max(11, 2) = 11
 \end{aligned}$$

$$d((12), 4) = \max(d(1, 4), d(2, 4)) = \max(5, 3) = 5$$

next join 3, 4 into a cluster

$$\begin{array}{c} (12) \quad (34) \\ (12) \begin{bmatrix} (12) & (34) \\ 0 & \\ 11 & 0 \end{bmatrix} \end{array} \quad d((12), (34)) = \max(11, 5) = 11$$

∴ the clusters join as follows

$$1, 2, 3, 4 \rightarrow (12), 3, 4 \rightarrow (12), (34) \rightarrow (1234)$$

(ii) in average linkage the distance between 2 clusters is given by

$$d(A, B) = \frac{\sum_{j \in B} \sum_{i \in A} d(i, j)}{n_A \times n_B}$$

where n_A is the no. of elements in cluster A

$$\begin{matrix} & 1 & 2 & 3 & 4 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{matrix} \xrightarrow{\text{join } 1, 2} \begin{matrix} & (12) & 3 & 4 \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{matrix}$$

$$d((12), 3) = \frac{d(1, 3) + d(2, 3)}{2} = \frac{11 + 2}{2} = 6.5$$

$$d((12), 4) = \frac{d(1, 4) + d(2, 4)}{2} = \frac{5 + 3}{2} = 4$$

next join 3, 4 into a cluster

$$\begin{matrix} & (12) & (34) \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{matrix} \quad d((12), (34)) = \frac{d(1, 3) + d(1, 4) + d(2, 3) + d(2, 4)}{4} = \frac{11 + 2 + 3 + 5}{4} = \frac{21}{4} = 5.25$$

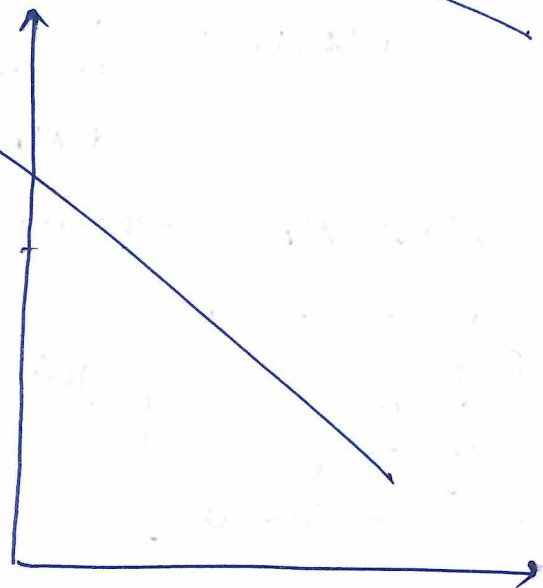
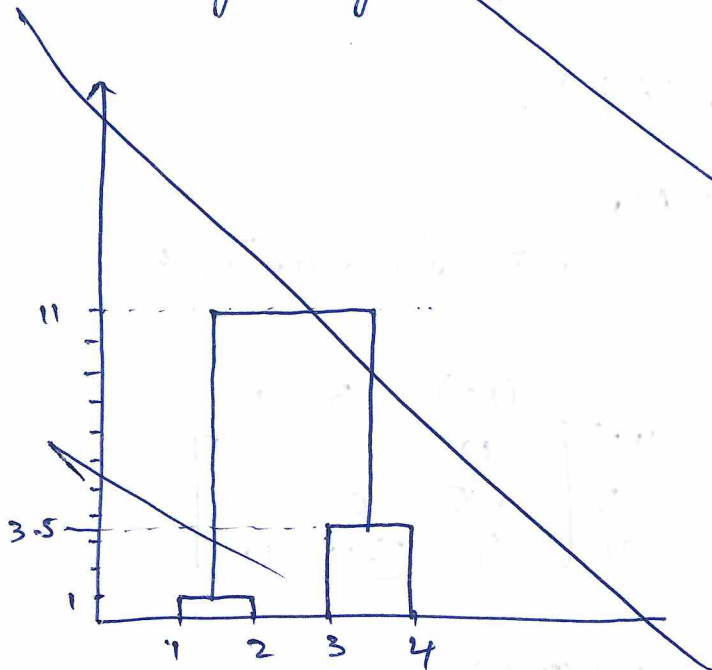
∴ the clusters form as follows

$$1, 2, 3, 4 \rightarrow (12), 3, 4 \rightarrow (12), (34) \rightarrow (1234)$$

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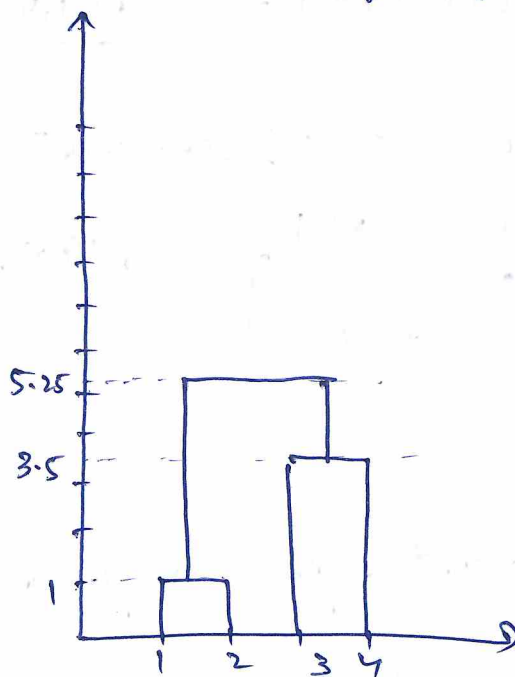
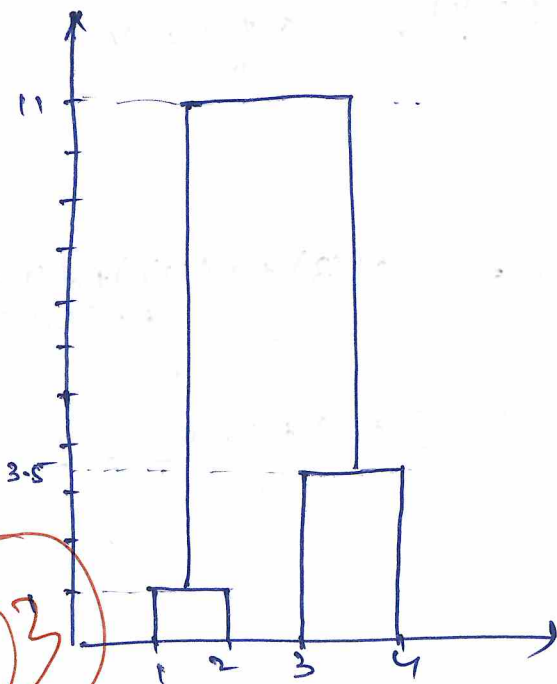
(b) dendrogram for (i)

dendrogram for (ii)



dendrogram for (i)

dendrogram for (ii)



03 It can be observed that in both the cases, the clusters (1,2) and (3,4) form around the same time however, since in complete linkage distance between (1,2) and (3,4) is 11, ^{as} ~~dis~~ compared to 5.25 in avg. linkage, the clusters (1,2,3,4) forms much later in (i) as compared to (ii)

Q2 part b)

$$P = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

the first principal component is given by $e_1' X_{\text{standard}}$
 where e_1 is the corresponding eigenvector to largest eigenvalue
 ~~X_{standard} is a standardised variables~~
 to find eigenvalues of $P \rightarrow$

$$\det \begin{pmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{pmatrix} = 0 \Rightarrow (1-\lambda) \left((1-\lambda)^2 - \frac{1}{2} \right) + \frac{1}{\sqrt{2}} \left(0 - \frac{1-\lambda}{\sqrt{2}} \right) = 0$$

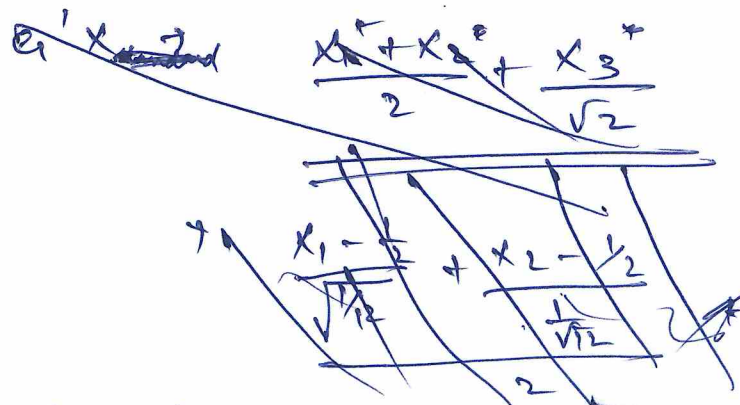
$$\Rightarrow (1-\lambda) \left[(1-\lambda)^2 - \frac{1}{2} - \frac{1}{2} \right] = 0$$

$$\Rightarrow \lambda(1-\lambda)(2-\lambda) = 0 \Rightarrow \lambda = 0, 1, 2$$

so $\lambda_1 = 2$, now corresponding eigenvector

$$\Rightarrow \begin{bmatrix} -1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \Rightarrow \begin{aligned} a &= \frac{c}{\sqrt{2}} \\ b &= \frac{c}{\sqrt{2}} \\ c &= \frac{a+b}{\sqrt{2}} \end{aligned} \Rightarrow e_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

so first principal component is $e_1' X = \frac{X_1 + X_2}{2} + \frac{X_3}{\sqrt{2}}$



variance of the first component is given by

$$\frac{1}{4} \text{Var}(X_1) + \frac{1}{4} \text{Var}(X_2) + \frac{1}{2} \text{Var}(X_3) + \frac{2\rho}{2\sqrt{2}} \text{Cov}(X_1, X_3) + \frac{2\rho}{2\sqrt{2}} \text{Cov}(X_2, X_3)$$

$$\Rightarrow \frac{1}{4} \times \frac{1}{12} + \frac{1}{4} \times \frac{1}{12} + \frac{1}{2} \times \frac{1}{6} + \frac{2 \times \frac{1}{\sqrt{2}}}{2\sqrt{2}} \times \frac{1}{12} + \frac{2 \times \frac{1}{\sqrt{2}}}{2\sqrt{2}} \times \frac{1}{12}$$

$$\Rightarrow \frac{1}{24} + \frac{1}{12} + \frac{1}{24} + \frac{1}{24} \Rightarrow \frac{1}{8} + \frac{1}{12} \Rightarrow \underline{\underline{\frac{2}{16}}}$$

~~\therefore~~ Variance of first principal component is $\frac{2}{16}$

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